



59 CONGRESO SEPAR

4-5-6 JUNIO 2026
A CORUÑA

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INSTITUTO DE INVESTIGACIÓN BIOSANITARIA

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Enfermedades Respiratorias



Tapones mucosos en la EPOC

¿La terapia biológica tiene papel?

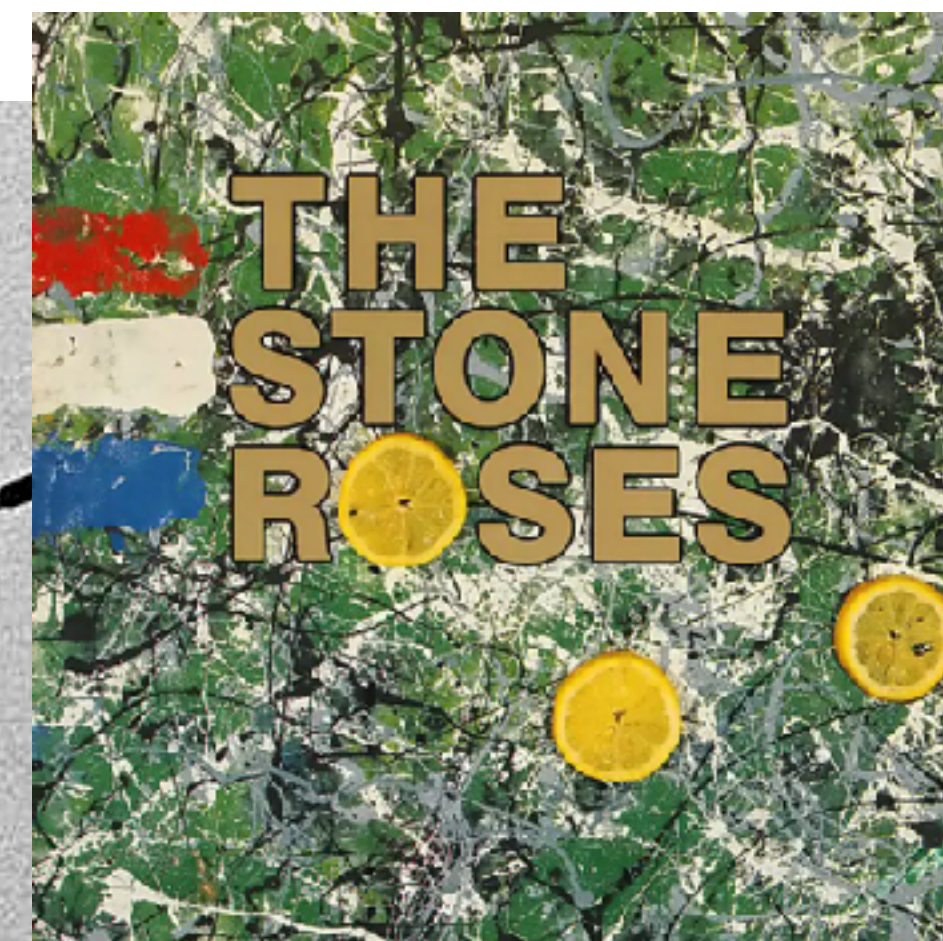
B. Alcazar Navarrete

Jefe de Sección de Neumología

Universidad de Granada

H.U. Virgen de las Nieves. Granada

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Nick Abbot

10pm - 1am

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Non-stop News

11pm - 6am



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20 Nov 2025, 16:16 |
Updated: 10 Feb 2026

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The Stone Roses legend Gary 'Mani' Mounfield dies aged 63 as tributes pour in

The musician is understood to have died at his home in Heaton Moor Stockport after collapsing.



Latest News

Liverpool to open talks with Andoni Iraola after sacking Arne Slot
4h ago



Dua Lipa and Callum Turner tie the knot in intimate London wedding ceremony
4h ago



Travel chaos looms: Last-gasp talks to avert London Underground strikes next week
5h ago



Search launched for boy, 11, last seen entering South Yorkshire river after 14 Brits across UK drown in open water
5h ago



Manchester City secure domestic double with Women's FA Cup victory against



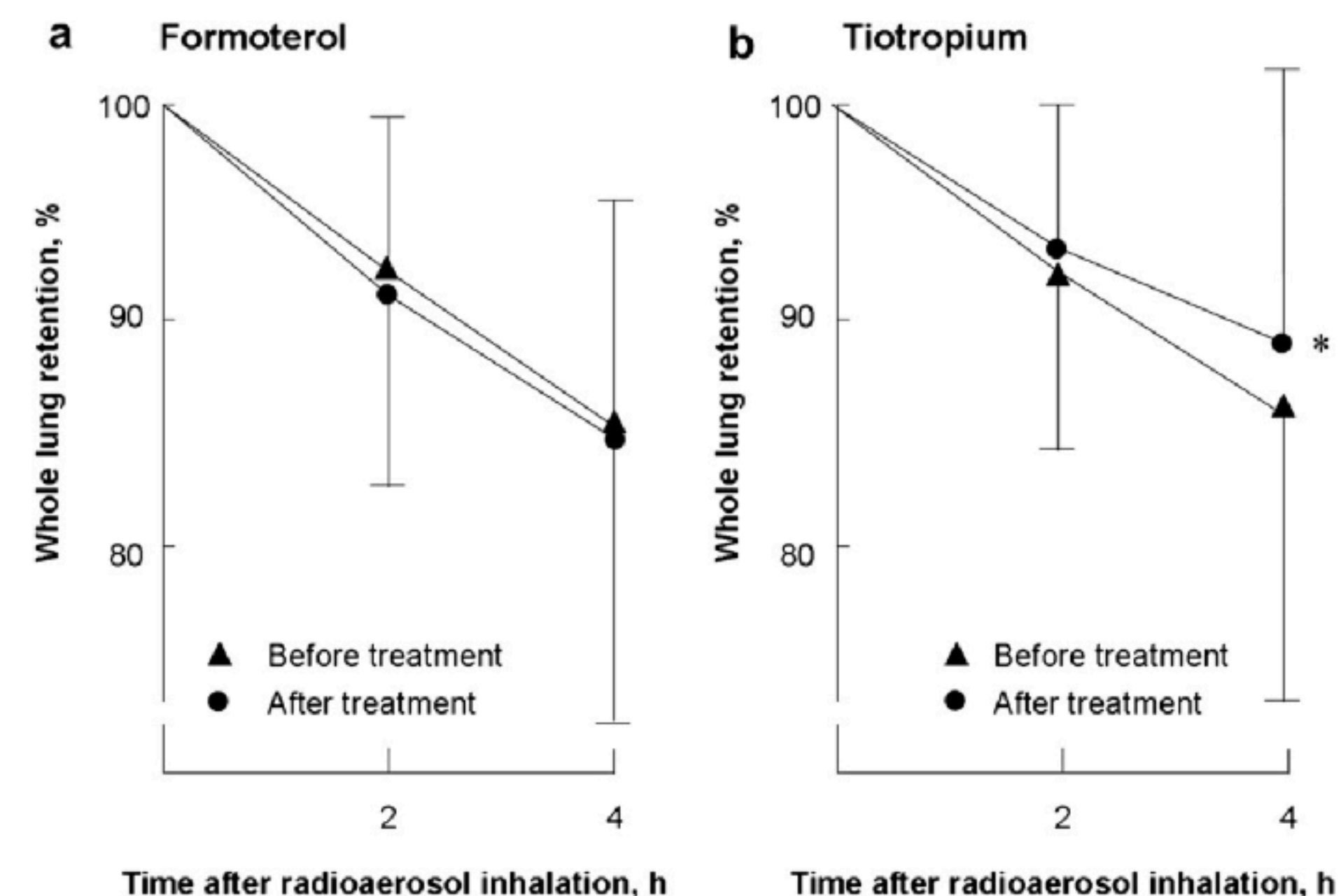
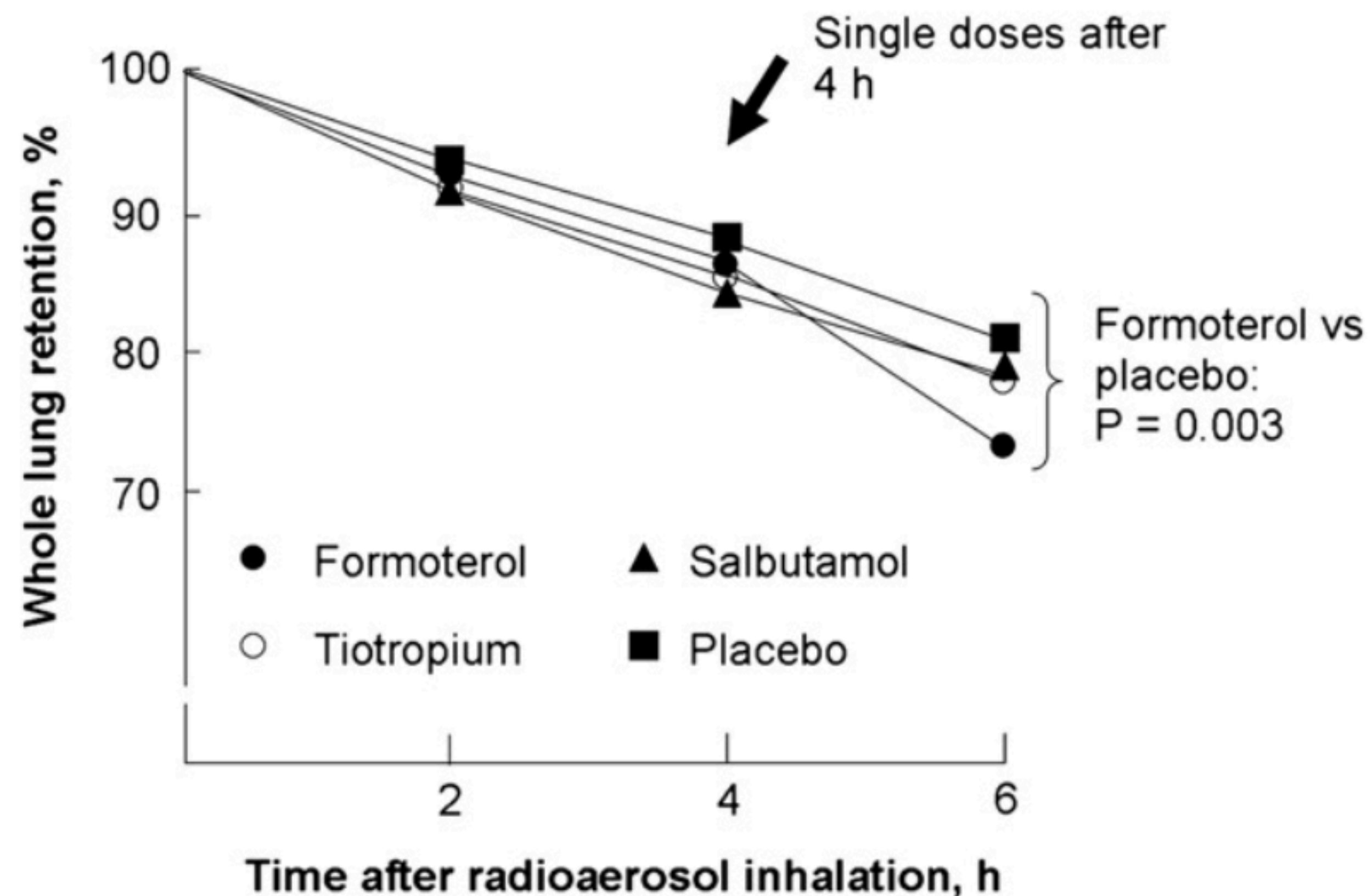
Conflictos de intereses (2023- 2026)

- Pagos por conferencias/ asistencia a congresos/ asesoría científica:
 - AstraZeneca/ Bial/ Chiesi/ CSL/ Grifols/ GSK/ MSD/ PulmonX/ Sanofi/ Zambon
- Acciones o relaciones familiares con industria farmacéutica: NO
- Relación con industria del tabaco: NO
- Participación en guías de práctica clínica: Comité Científico de GesEPOC
- Patentes: P201730724 (registrada)

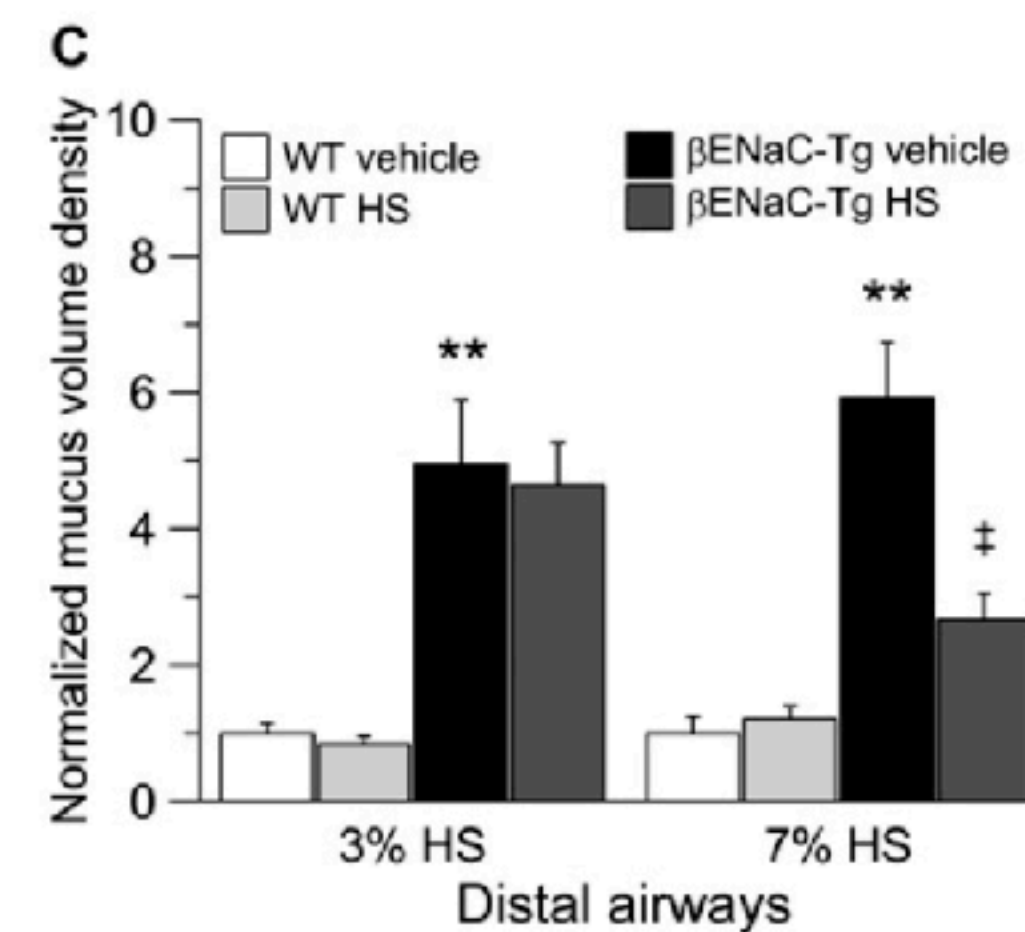
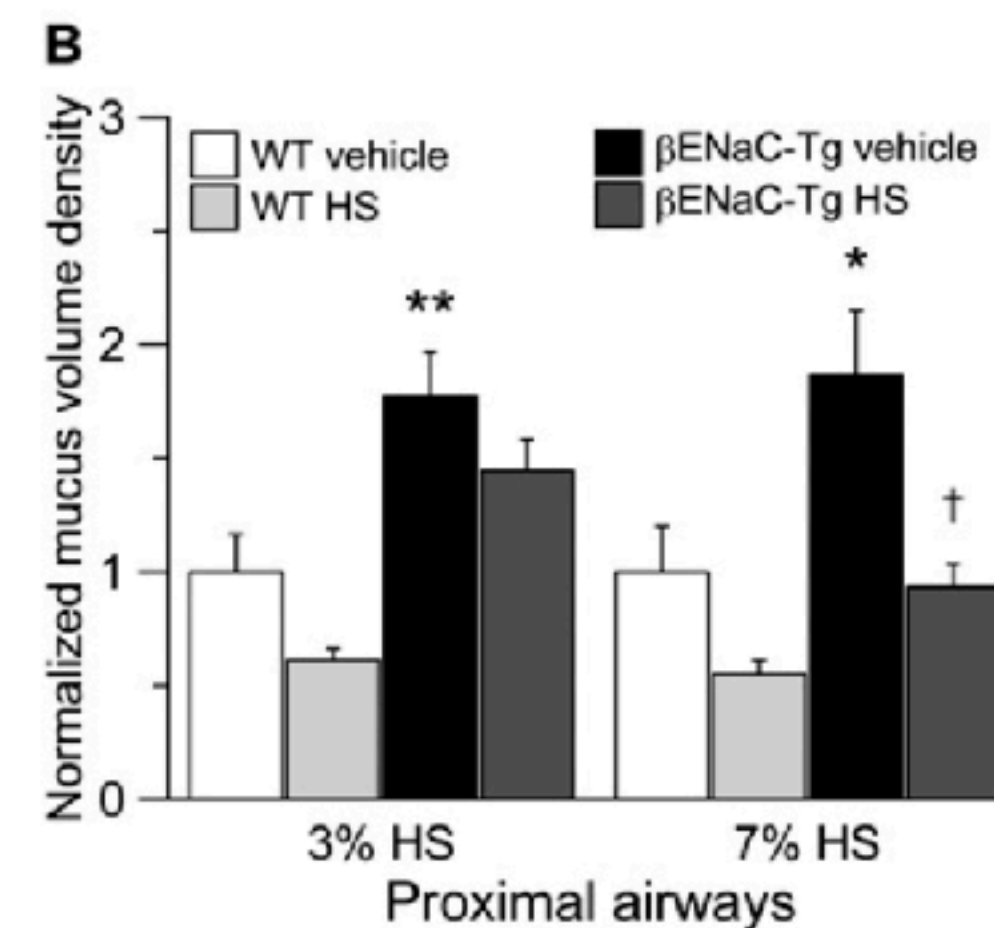
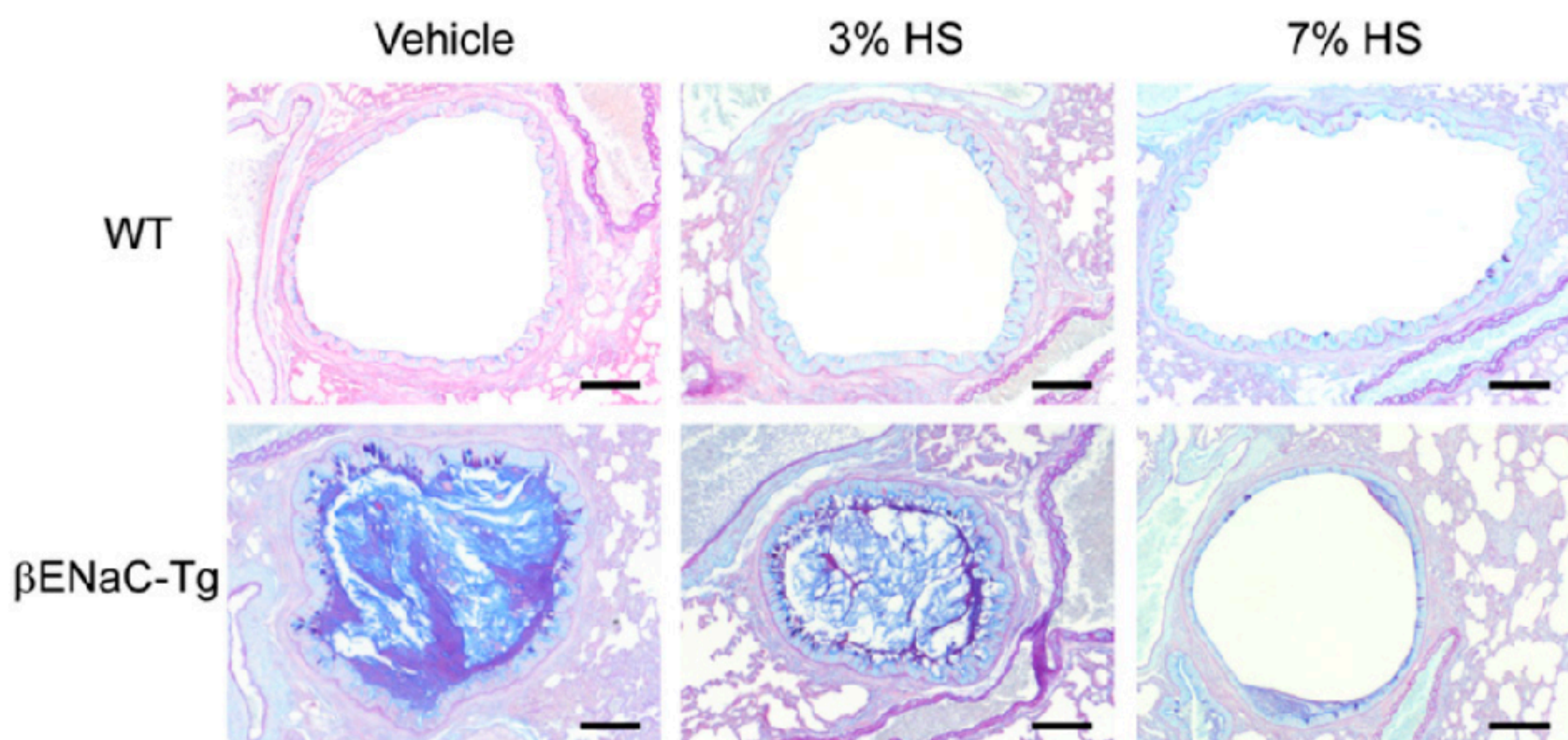
Agenda

- Tratamientos inhalados y su efecto en tapones de moco
- Mucolíticos y fármacos orales
- Dupilumab: evidencia disponible y estudios en marcha.
- Tozorakimab: evidencia disponible y estudios en marcha.
- Consideraciones finales

Fármacos inhalados



Fármacos inhalados- SSH

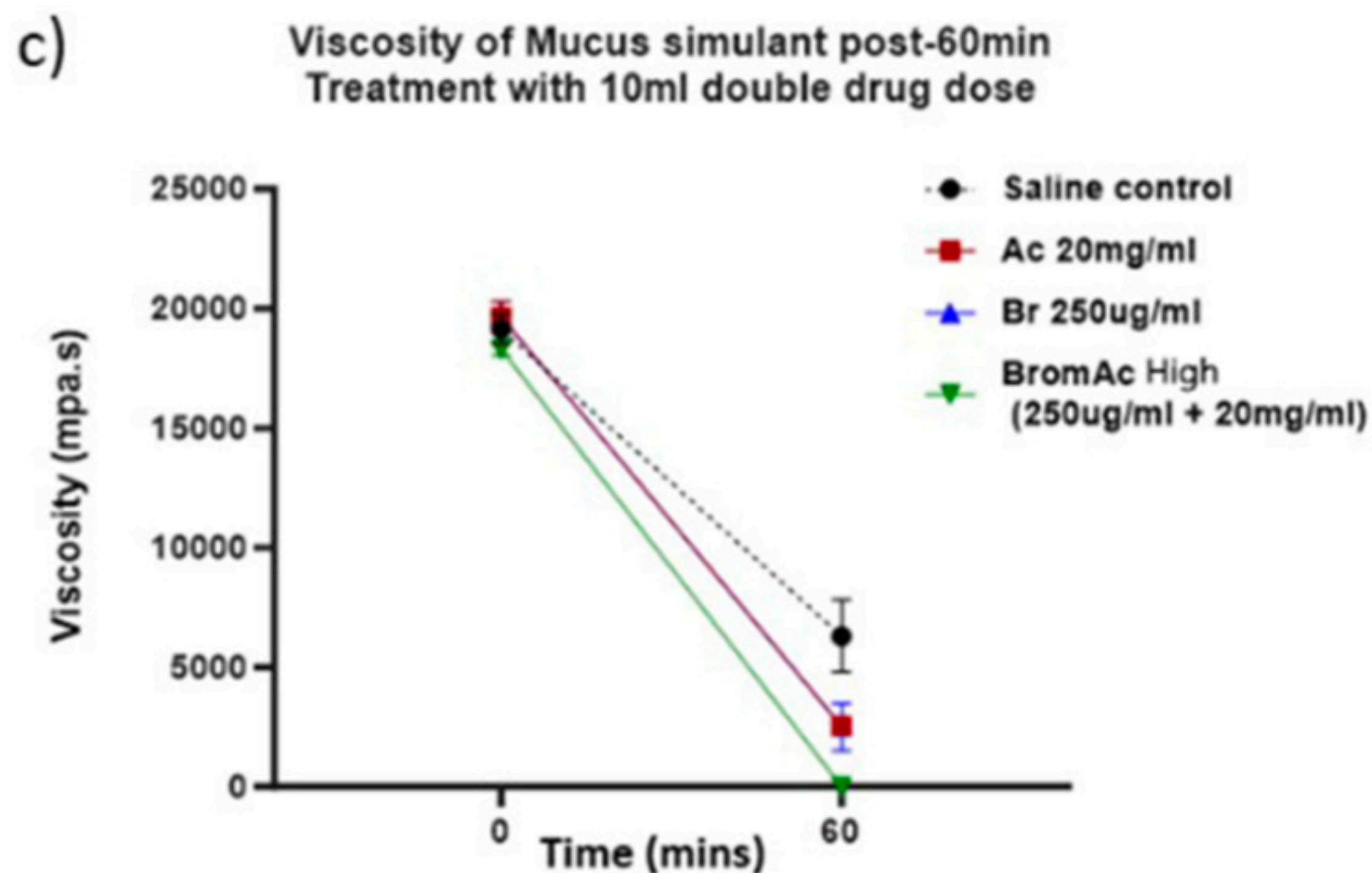
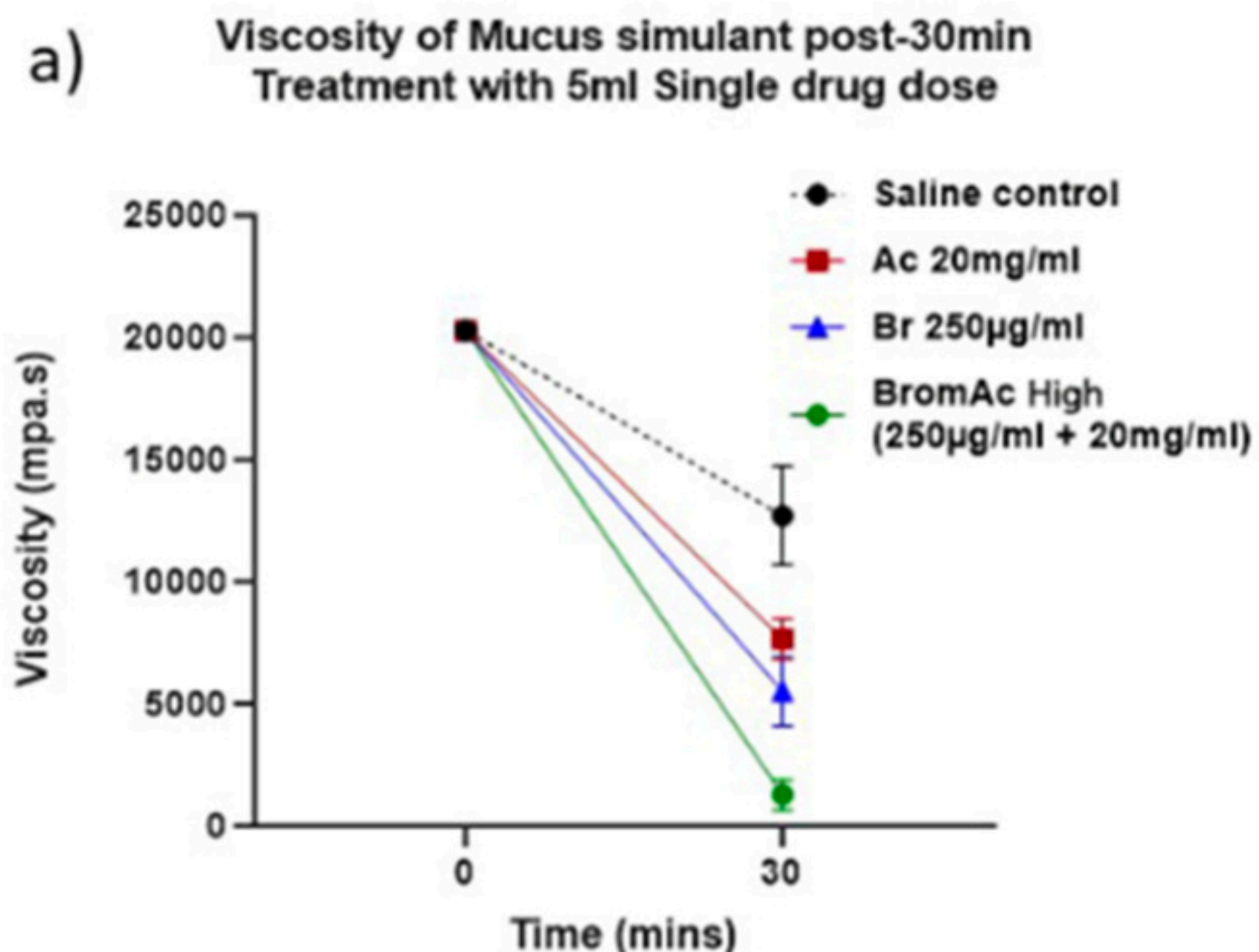


Graeber SY, Zhou-Suckow Z, Schatterny J, Hirtz S, Boucher RC, Mall MA. Hypertonic saline is effective in the prevention and treatment of mucus obstruction, but not airway inflammation, in mice with chronic obstructive lung disease. *Am J Respir Cell Mol Biol.* 2013;49(3):410-417. doi:10.1165/rcmb.2013-0050OC

Agenda

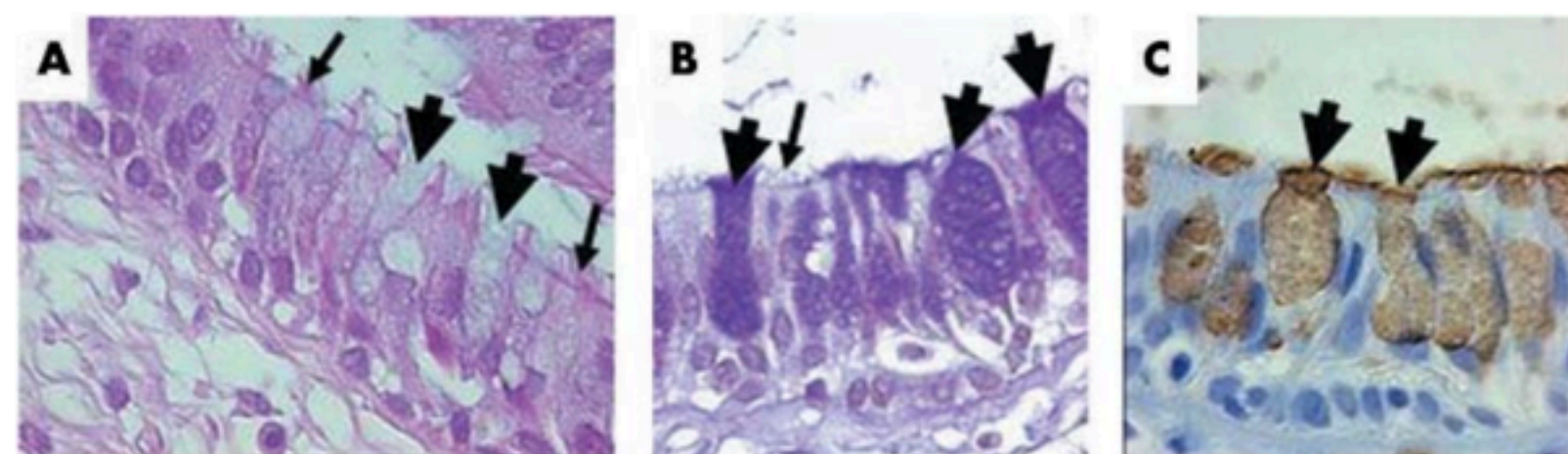
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Mucolíticos

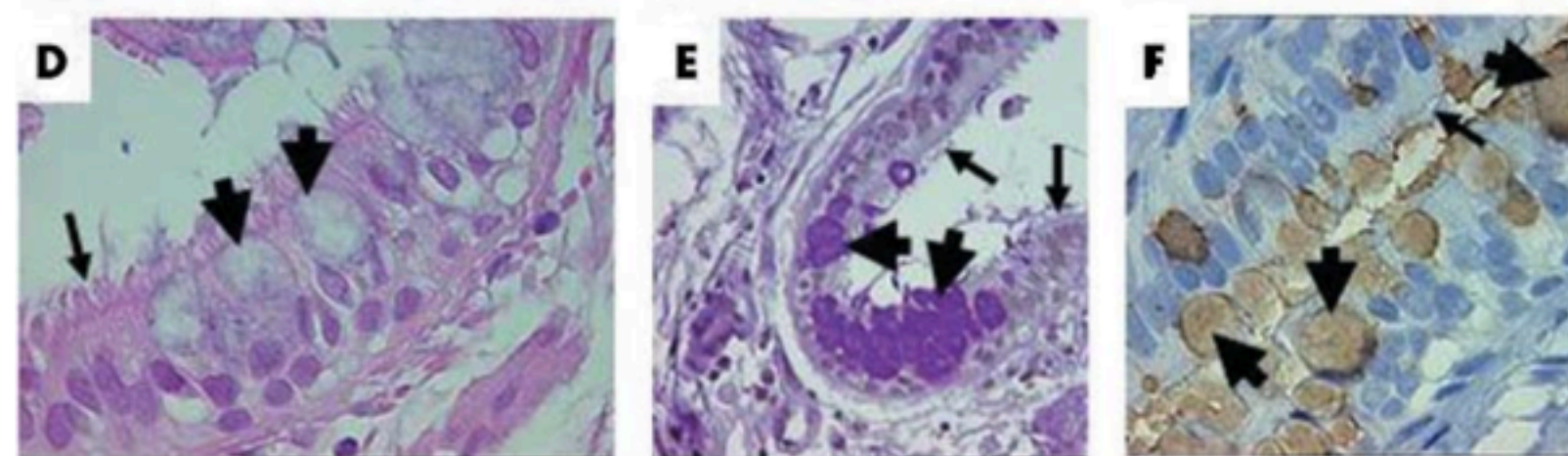


iPDE4- Roflumilast

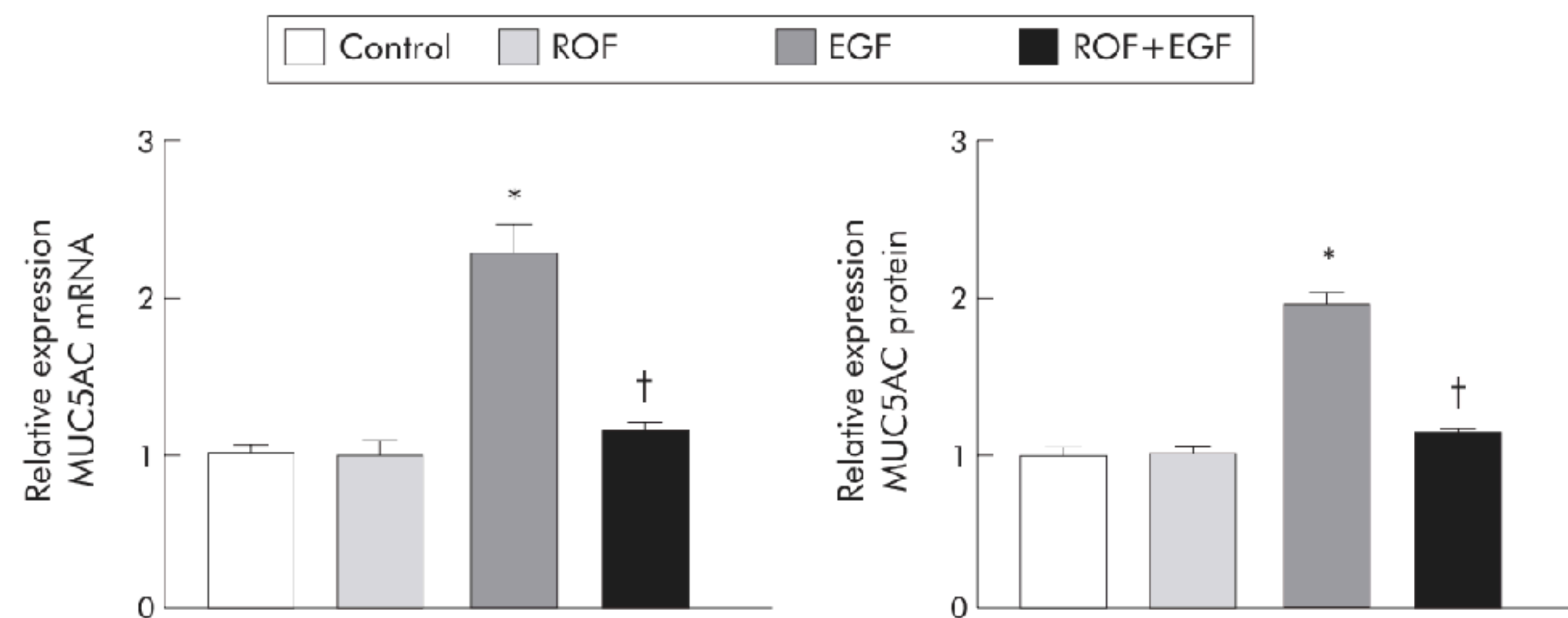
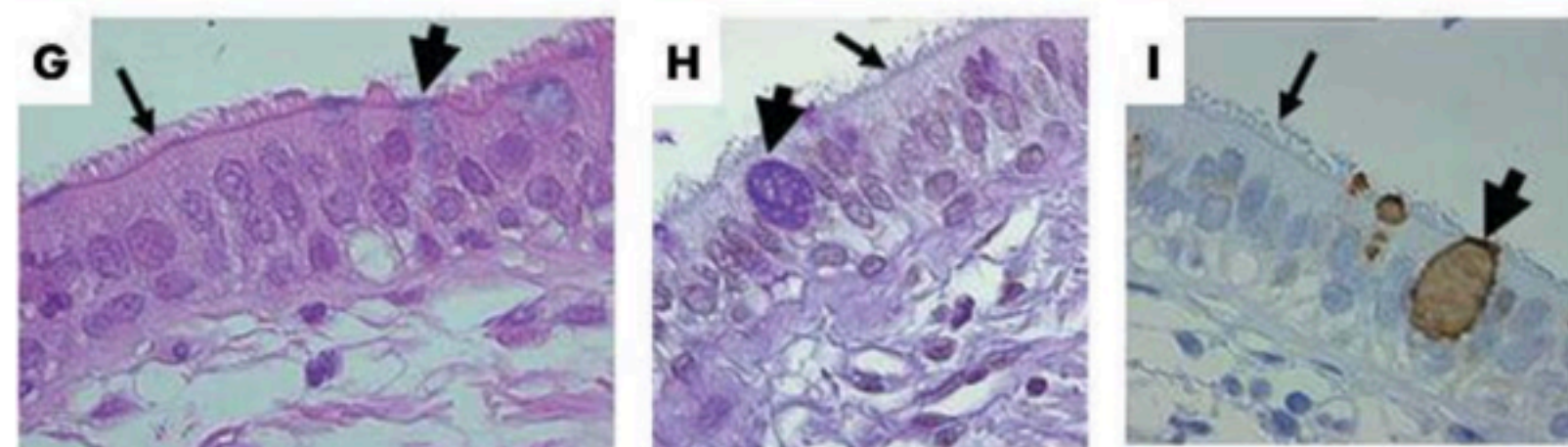
Control



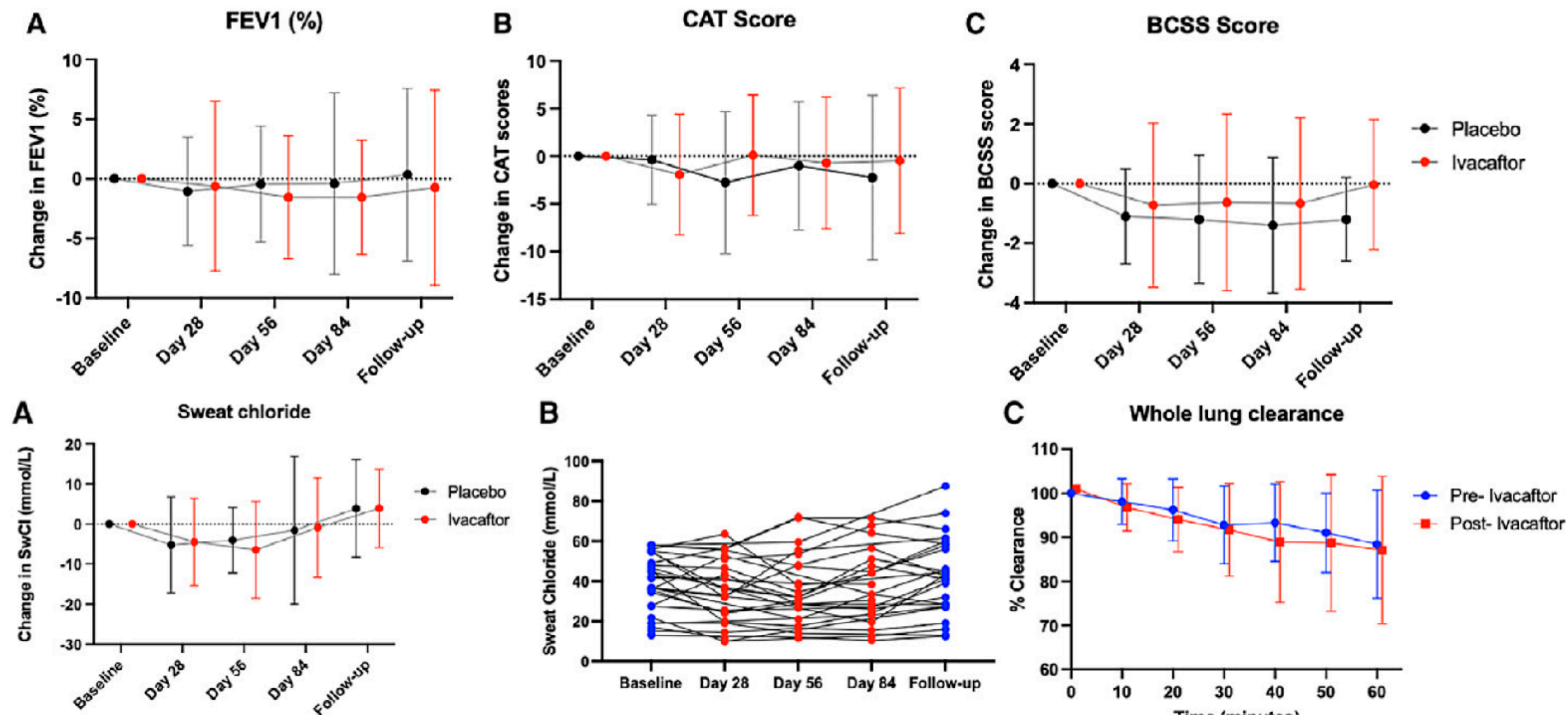
EGF



iPDE4



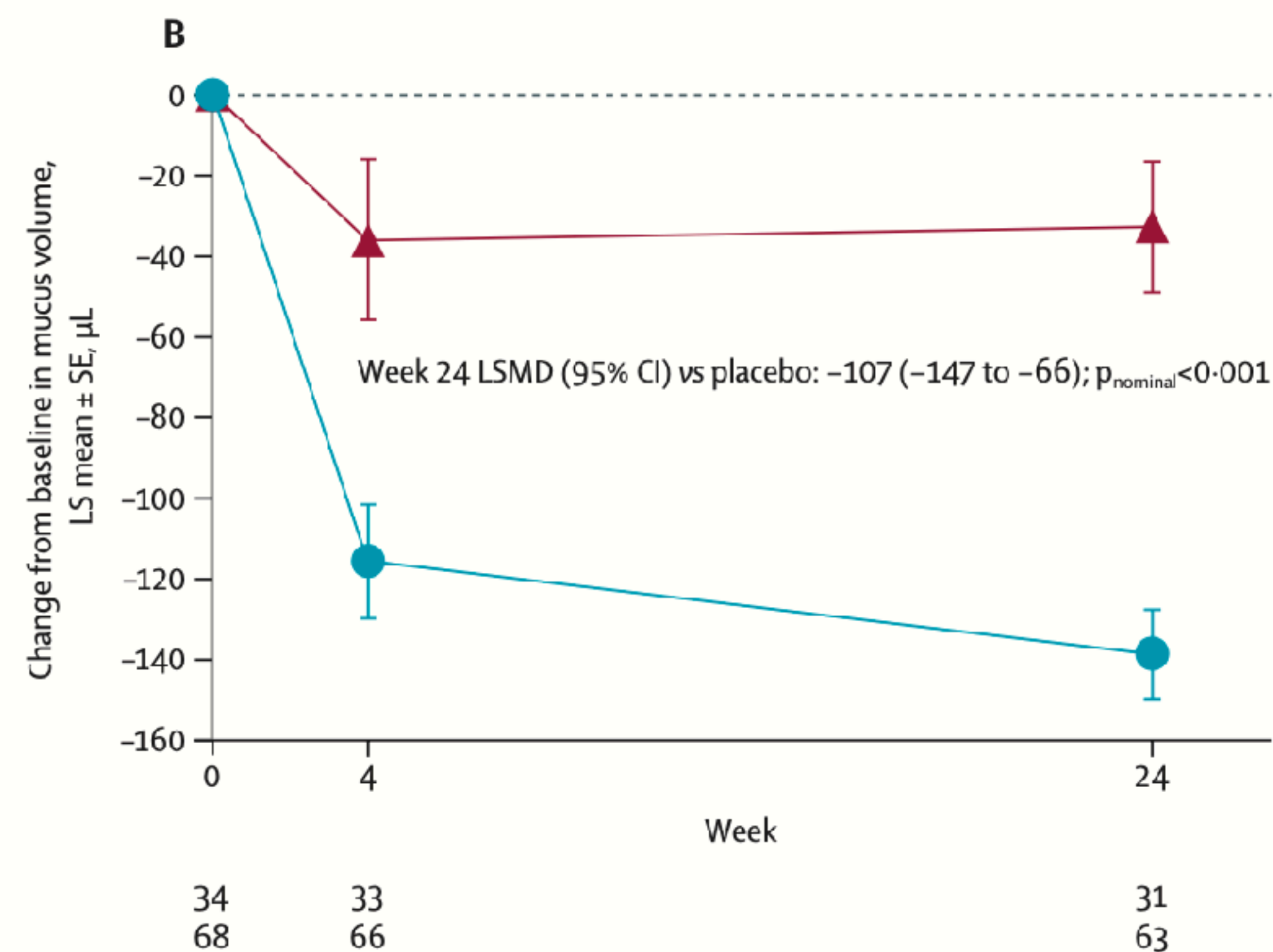
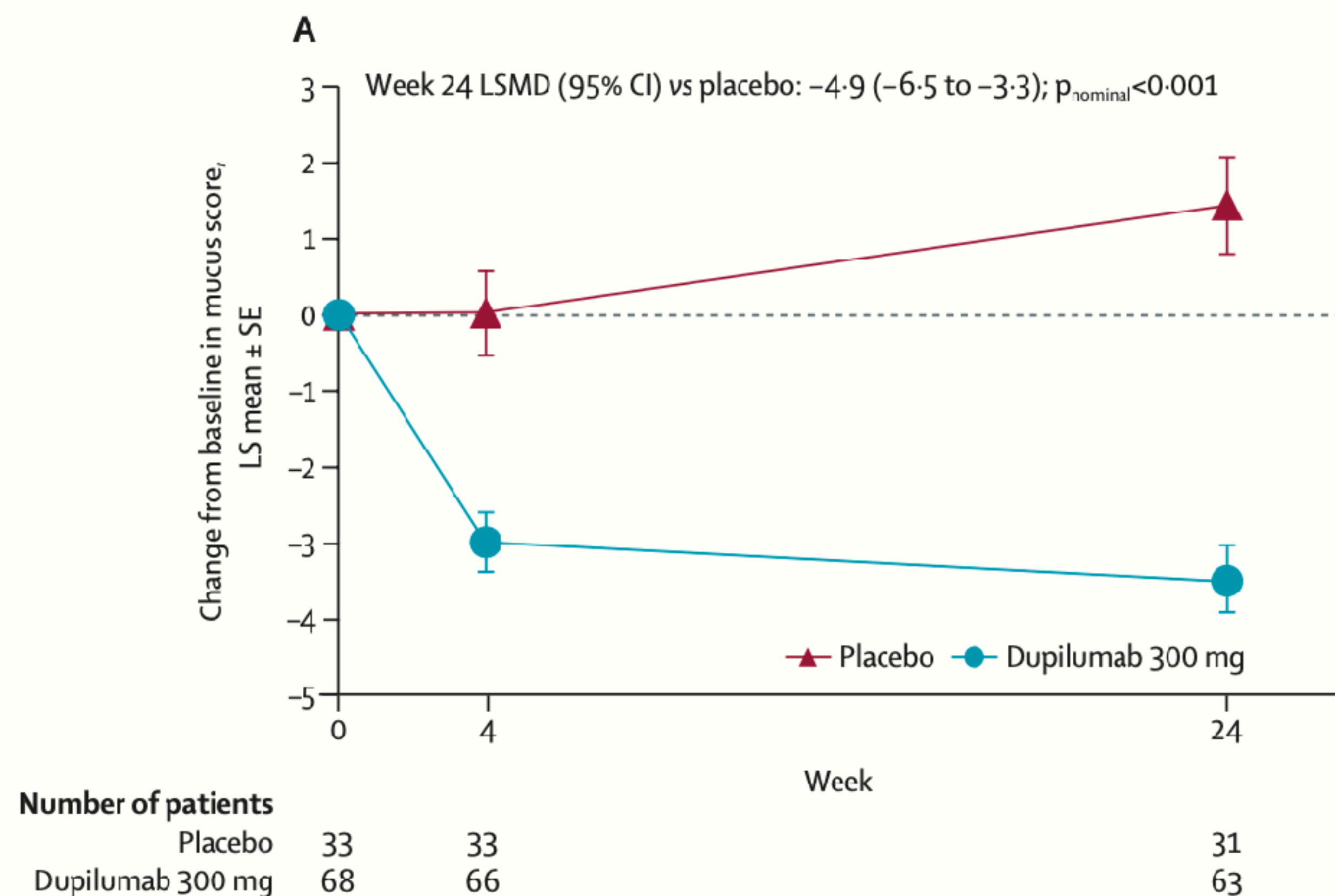
Potenciadores del CFTR- Ivacaftor



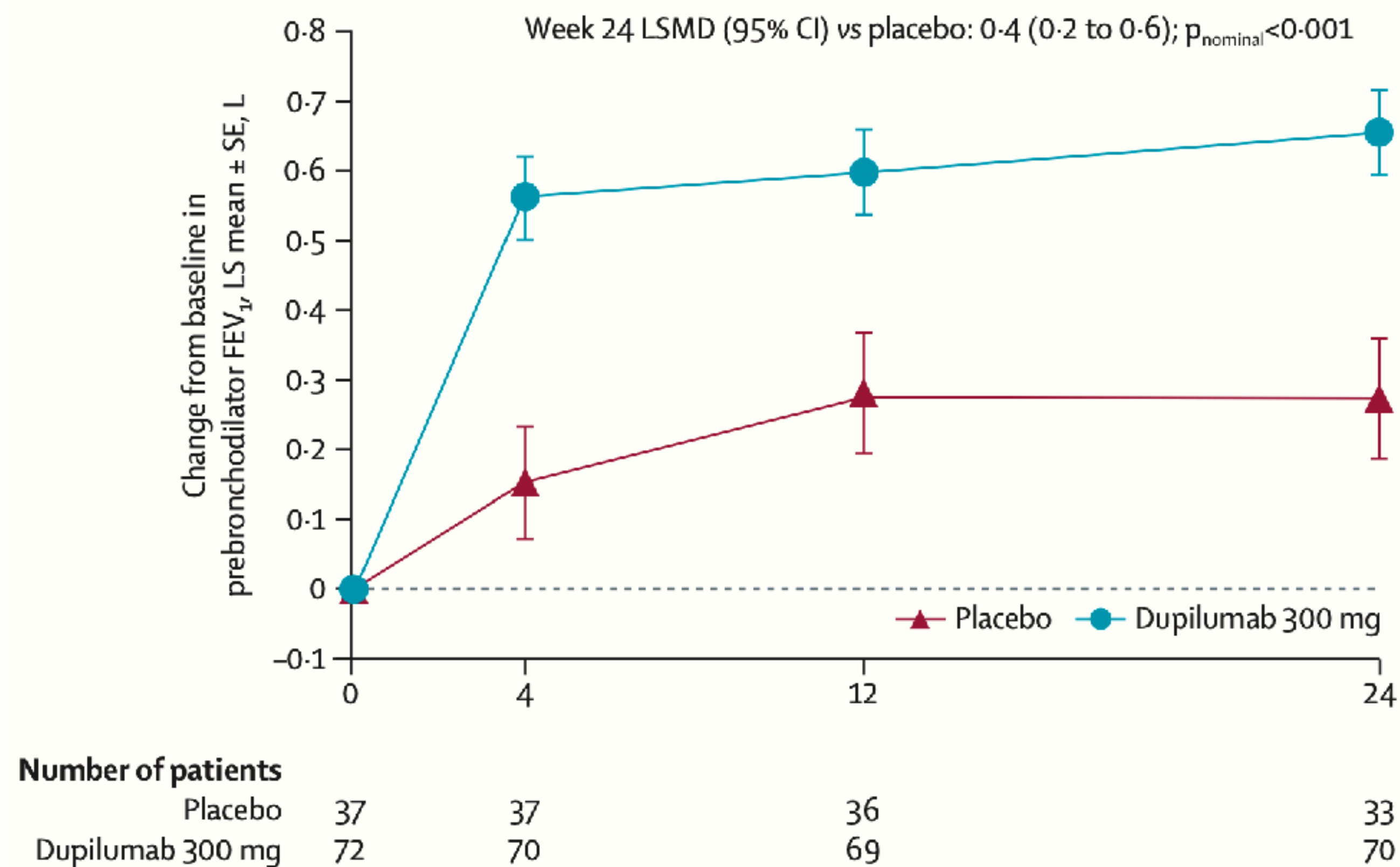
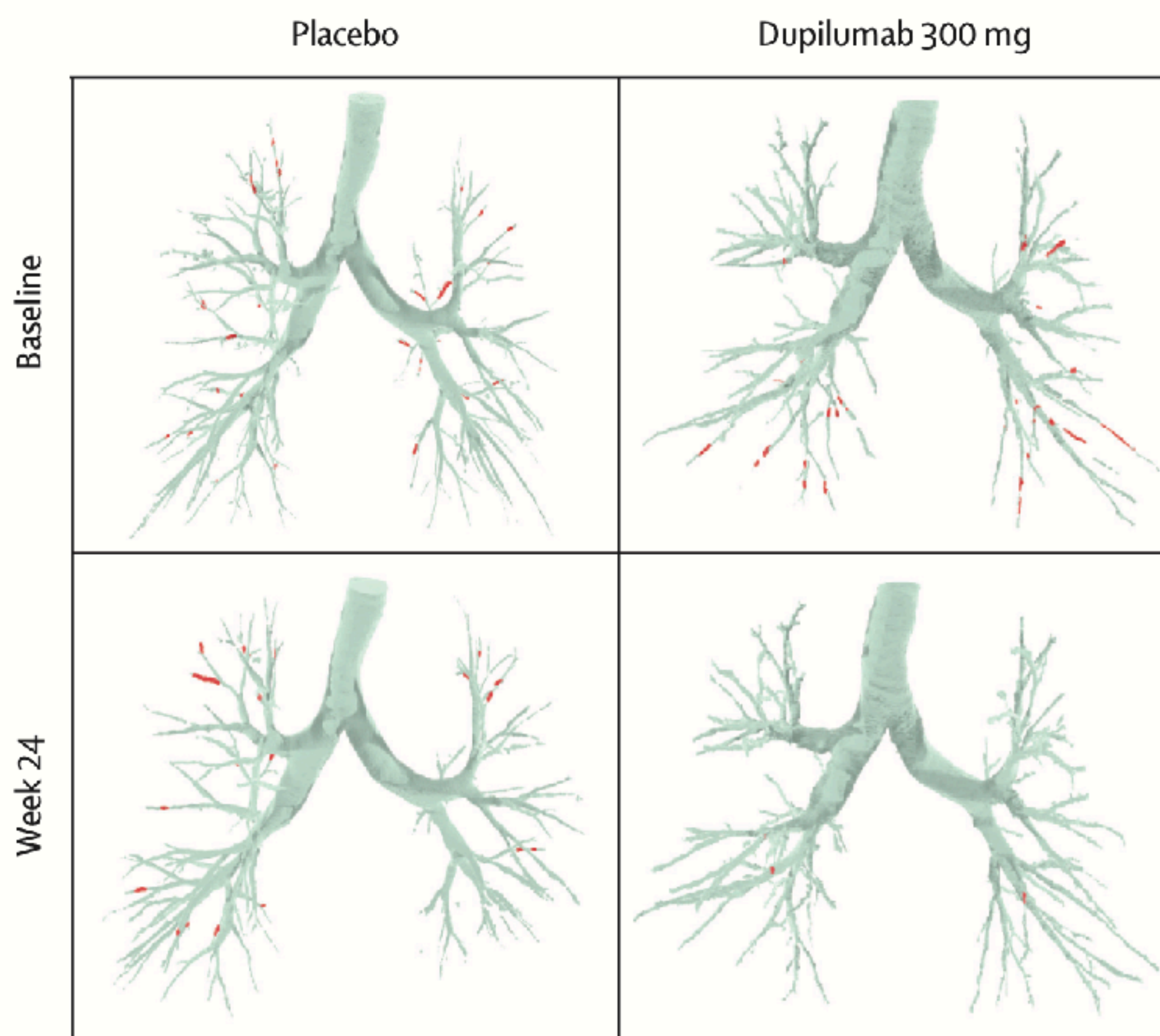
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Dupilumab- Estudio VESTIGE



Dupilumab- Estudio VESTIGE



Dupilumab

Design of the Phase 4 AEOLUS Study Assessing the Effect of Dupilumab on Airway Inflammation and Remodeling through Lung Imaging Parameters in Patients with COPD

Surya P. Bhatt¹, Alejandro A. Diaz², George R. Washko³, Jadwiga A. Wedzicha³, Grace Parraga⁴, Christopher E. Brightling⁵, Mei Zhang⁶, Mena Soliman⁷, Danen M. Cunoosamy⁸, Parameswaran Nair^{9,10}, Sarah Svenningsen^{9,10}

¹University of Alabama at Birmingham, Birmingham, AL, USA; ²Brigham and Women's Hospital, Boston, MA, USA; ³National Heart and Lung Institute, Imperial College London, London, UK; ⁴Robarts Research Institute, The University of Western Ontario, London, ON, Canada; ⁵University of Leicester, Leicester, UK; ⁶Sanofi, Morristown, NJ, USA; ⁷Regeneron Pharmaceuticals Inc., Tarrytown, NY, USA; ⁸Sanofi, Cambridge, MA, USA; ⁹McMaster University, Hamilton, ON, Canada; ¹⁰Firestone Institute for Respiratory Health, St. Joseph's Healthcare Hamilton, Hamilton, ON, Canada

Objective

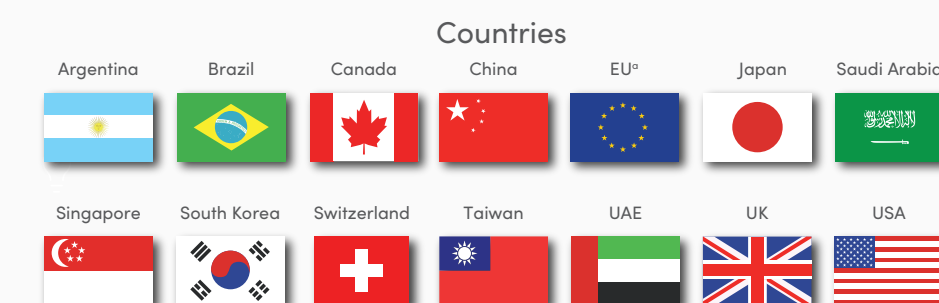
To evaluate the effect of dupilumab on airway resistance and remodeling, including mucus plugging, airway inflammation, and their association with improvements in exacerbations, lung function, symptoms, and quality of life in patients with COPD and type 2 inflammation

Background

- IL-4 and IL-13, key and central drivers of type 2 inflammation, contribute to mucus hypersecretion and mucus plug formation via promotion of goblet cell differentiation, excess mucin (MUC5AC) production, and associated mucociliary disruption, leading to poorer disease outcomes in patients with COPD¹⁻⁵
- Dupilumab can reduce mucus plugs and improve small airway function in patients with moderate-to-severe asthma, as shown in the VESTIGE trial⁶
- The phase 3 BOREAS and NOTUS trials demonstrated that dupilumab reduces moderate or severe exacerbations and improves lung function, symptoms, and quality of life in patients with COPD and type 2 inflammation⁷⁻¹⁰
 - However, currently, there are limited data on the effect of dupilumab on mucus plugging in patients with COPD

Methods

- AEOLUS (NCT07053423) is a phase 4, randomized, double-blind, placebo-controlled, parallel-group, multinational trial that will enroll patients aged 40 to 85 years with moderate or severe COPD
- Approximately 218 patients from 23 countries across Asia, Europe, Latin America, and North America will be randomized (2:1) to receive subcutaneous add-on dupilumab 300 mg or placebo q2w for 24 weeks



*Austria, Denmark, France, Germany, Hungary, Italy, Netherlands, Poland, Spain, Sweden.

Results

Figure 1. Study design of the phase 4 AEOLUS clinical trial

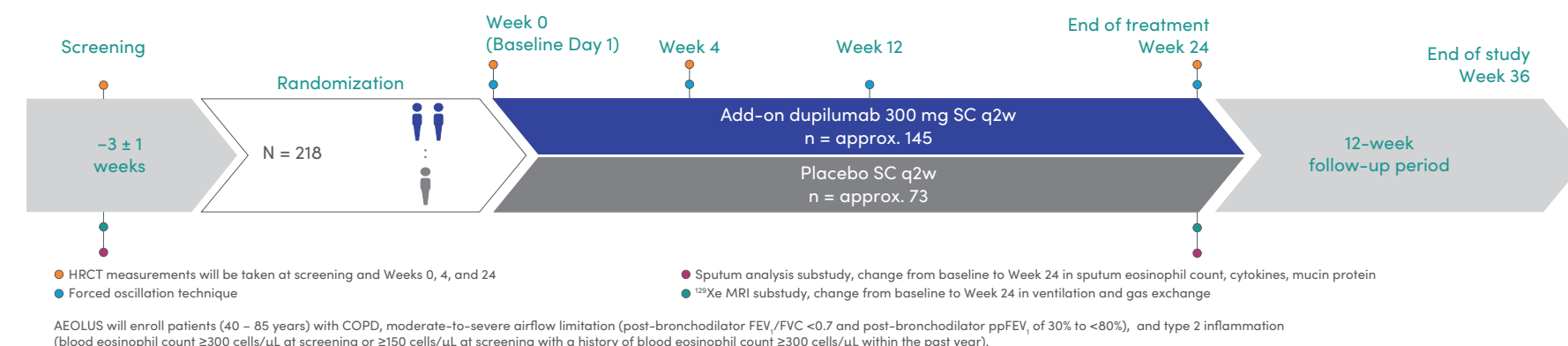


Table 1. Key patient eligibility criteria

Inclusion criteria

- 40 to 85 years of age**
- BMI of 16 to 40 kg/m²**
- Former smokers** (history of ≥10 pack-years)
Current smokers (stopped for ≥6 months before the screening visit)
- Physician-diagnosed COPD for ≥1 year** with post-bronchodilator FEV₁/FVC ratio <0.70 and post-bronchodilator ppFEV₁ ≥30% and <80%
- Exacerbation history:** ≥2 moderate^a or ≥1 severe^b exacerbation in the year prior
- Inhaled triple therapy (ICS/LABA/LAMA) for ≥3 months before randomization** was permitted if ICS was contraindicated, with a stable dose of medication for ≥1 month before the screening visit; inhaled dual therapy
- Blood eosinophil count ≥300 cells/μL at screening or ≥150 cells/μL at screening and a history of ≥300 cells/μL within the past year during stable state**
- MRC Dyspnea Scale grade ≥2 or CAT score ≥10 at screening**
- CT mucus plug score ≥3 at screening**

^aDefined as an exacerbation requiring treatment with either systemic corticosteroids or antibiotics. 1 of the 2 required moderate exacerbations must require treatment with systemic corticosteroids to fulfill the inclusion criterion. ^bDefined as an exacerbation requiring hospitalization or observation for at least 24 hours in the emergency department or urgent care facility. ^cIncluding but not limited to renal, hepatic, cardiovascular, metabolic, neurologic, hematologic, ophthalmologic, respiratory, gastrointestinal, cerebrovascular disease, malignancies, autoimmune disease, tuberculosis, parasitic (helminth) infection, hepatitis B/C infection, HIV 1/2 infection, or substance and/or alcohol use disorder. ^dIncluding but not limited to lung fibrosis, sarcoidosis, interstitial lung disease, pulmonary hypertension, bronchiectasis, or Churg-Strauss syndrome.

AES, adverse event of special interest; AX, reactance area; BMI, body mass index; CAT, COPD Assessment Test; COPD, chronic obstructive pulmonary disease; CT, computed tomography; ECG, electrocardiogram; FeNO, fractional exhaled nitric oxide; FEV₁, forced expiratory volume in 1 second; FOT, forced oscillation technique; FVC, forced vital capacity; GINA, Global Initiative for Asthma; HRCT, high-resolution computed tomography; ICS, inhaled corticosteroid(s); IL, interleukin; LABA, long-acting β₂-agonist(s); LAMA, long-acting muscarinic antagonist(s); MRI, magnetic resonance imaging; pp, percent predicted; q2w, every 2 weeks; R5-R20, difference in airway resistance at 5 Hz and 20 Hz; SAE, serious adverse event; SC, subcutaneous; TEAE, treatment-emergent adverse event; TLC, total lung capacity; VDP, ventilation defect percent.

Conclusion

The AEOLUS trial will evaluate the effect of dupilumab on mucus plugging and airway remodeling using HRCT, FOT, and ¹²⁹Xe MRI techniques in patients with COPD and type 2 inflammation

P1493

COPD

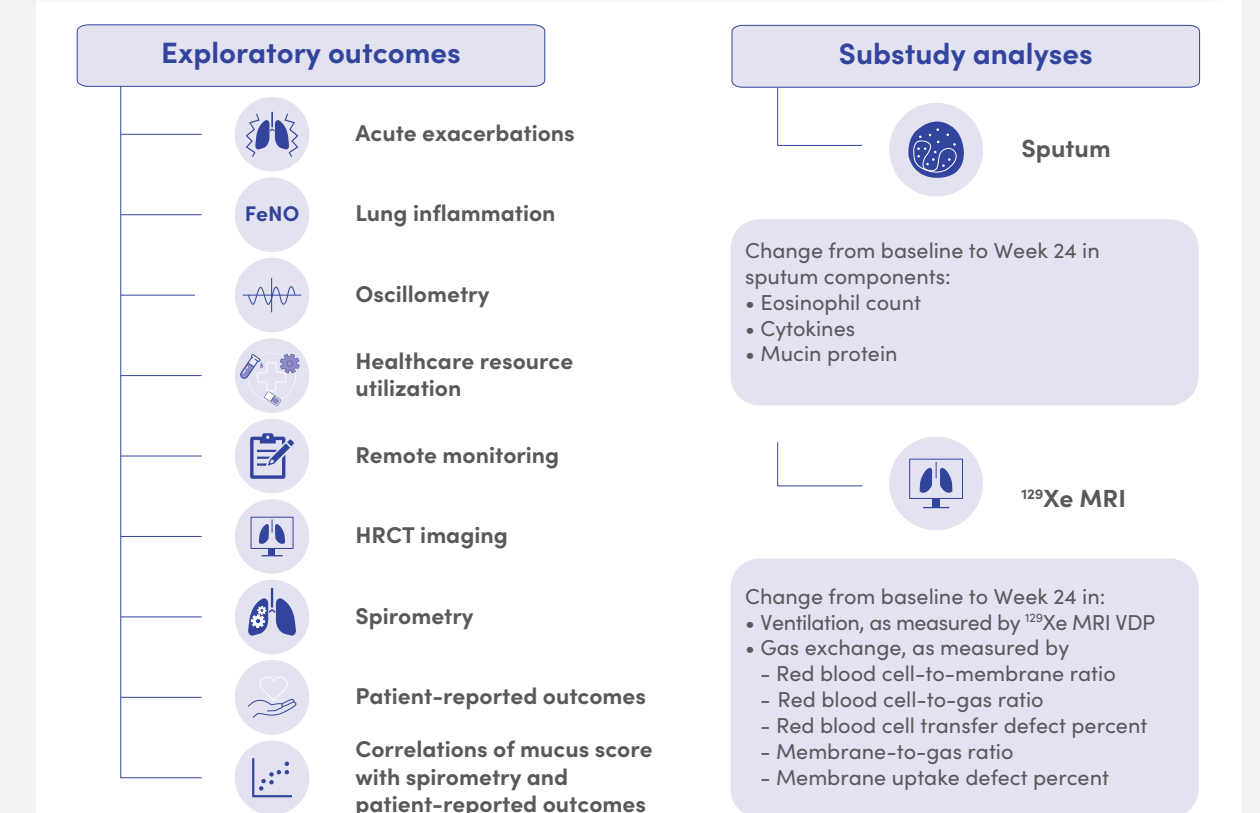


Full poster download
Copies of the presentation obtained through
the QR code are for personal reference only

Table 2. Summary of primary and secondary endpoints

Type of endpoint	Measurement	Endpoint
Primary		Change from baseline to Week 24 in global lung mucus score
Secondary	HRCT imaging	Change from baseline to Week 24 in global lung mucus volume and trimmed distal airway thickness at TLC
	Oscillometry	Change from baseline to Week 24 in R5-R20 and AX
	Safety	Incidence of TEAEs, SAEs, AESIs, and potentially clinically significant abnormalities in hematology, biochemistry, and vital signs

Figure 2. Exploratory endpoints and substudy analyses



Dupilumab

Inclusion criteria



40 to 85 years of age



BMI of 16 to 40 kg/m²



Former smokers (history of ≥ 10 pack-years)
Current smokers (stopped for ≥ 6 months before the screening visit)



Physician-diagnosed COPD for ≥ 1 year with
post-bronchodilator FEV₁/FVC ratio < 0.70 **and**
post-bronchodilator ppFEV₁ $\geq 30\%$ and $< 80\%$



Exacerbation history:
 ≥ 2 moderate^a or ≥ 1 severe^b exacerbation in the year prior



Inhaled triple therapy (ICS/LABA/LAMA) for ≥ 3 months before randomization was
permitted if ICS was contraindicated, with a stable dose of medication for ≥ 1 month
before the screening visit; inhaled dual therapy



Blood eosinophil count ≥ 300 cells/ μ L at screening **or** ≥ 150 cells/ μ L at screening **and**
a history of ≥ 300 cells/ μ L within the past year during stable state



MRC Dyspnea Scale grade ≥ 2 or CAT score ≥ 10 at screening



CT mucus plug score ≥ 3 at screening

Exclusion criteria



Acute exacerbation or respiratory tract infection ≤ 4 weeks
before/during screening period



Current diagnosis or history of asthma as defined in GINA
guidelines¹¹



Prior history of clinically significant laboratory findings or
abnormal ECG, or medical/clinical conditions^c at screening



Significant pulmonary disease other than COPD^d or systemic
disease associated with **elevated peripheral eosinophil count**

Diagnosis of **α -1 anti-trypsin deficiency**



Previous use of dupilumab or other biologic or
immunosuppressive therapy

Controller therapy compliance $< 80\%$ during screening



Live attenuated vaccination within 4 weeks of screening or
during the trial

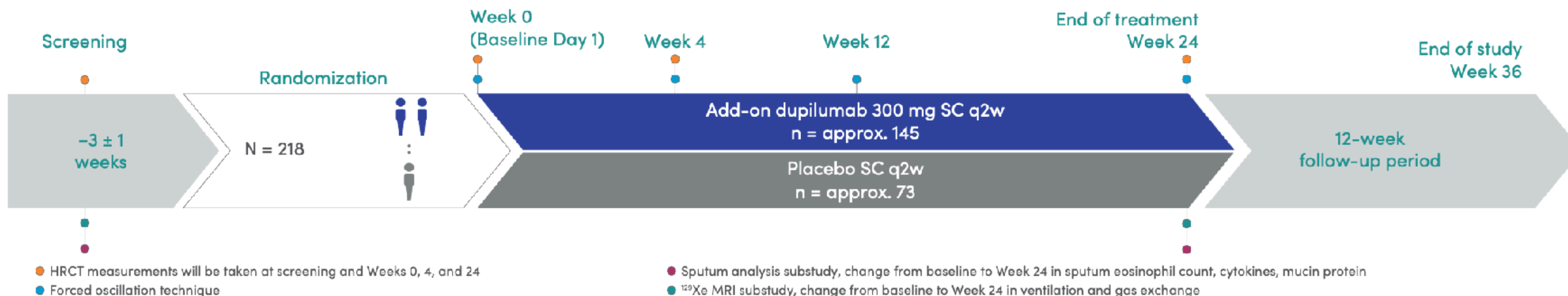


Treatment with **oxygen**



Unless on a stable dose for > 6 months, treatment with
antiarrhythmics, antidiuretics, macrolides, roflumilast,
ensifentrine, or mucolytics

Dupilumab



Type of endpoint	Measurement	Endpoint
Primary		Change from baseline to Week 24 in global lung mucus score
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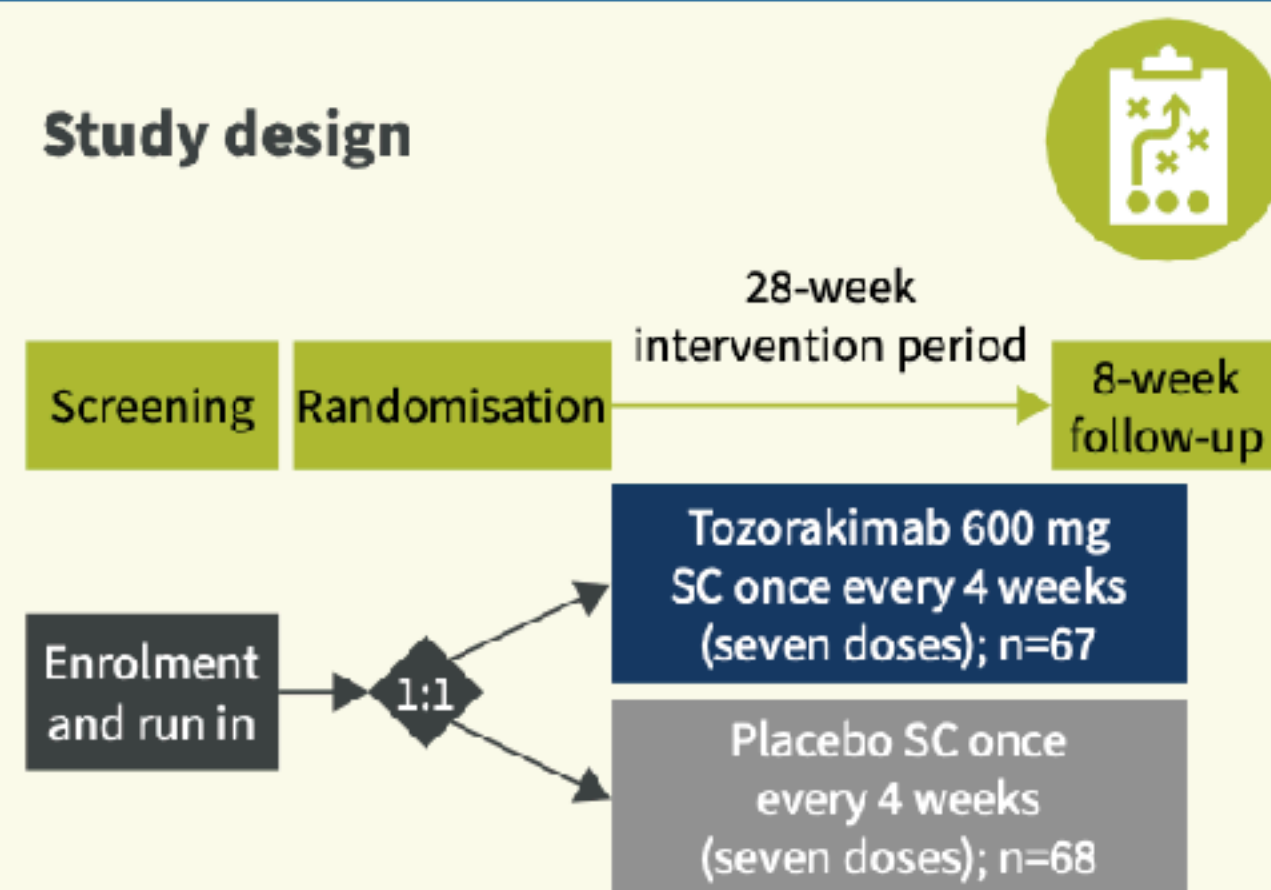
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Tozorakimab (anti- IL33)

A phase 2a trial of the IL-33 monoclonal antibody tozorakimab in patients with COPD: FRONTIER-4

Study design



Inclusion criteria

- Patients with COPD and chronic bronchitis
- Current or former smokers
- One or more exacerbations in the 24 months before screening

Primary outcome

LS mean difference *versus* placebo in change from baseline in **pre-BD FEV₁** at week 12

ITT population ↑ 24 mL
80% CI -15–63 mL;
p=0.216

Patients with at least two prior exacerbations ↑ 69 mL
80% CI 9–130 mL;
p=0.072

Secondary outcome

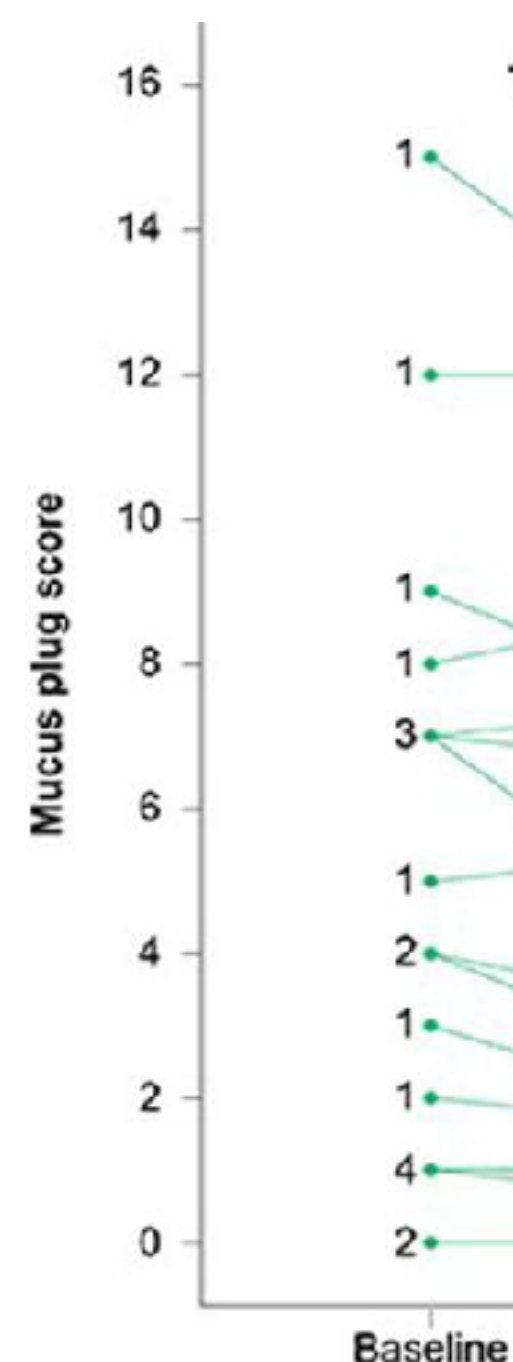
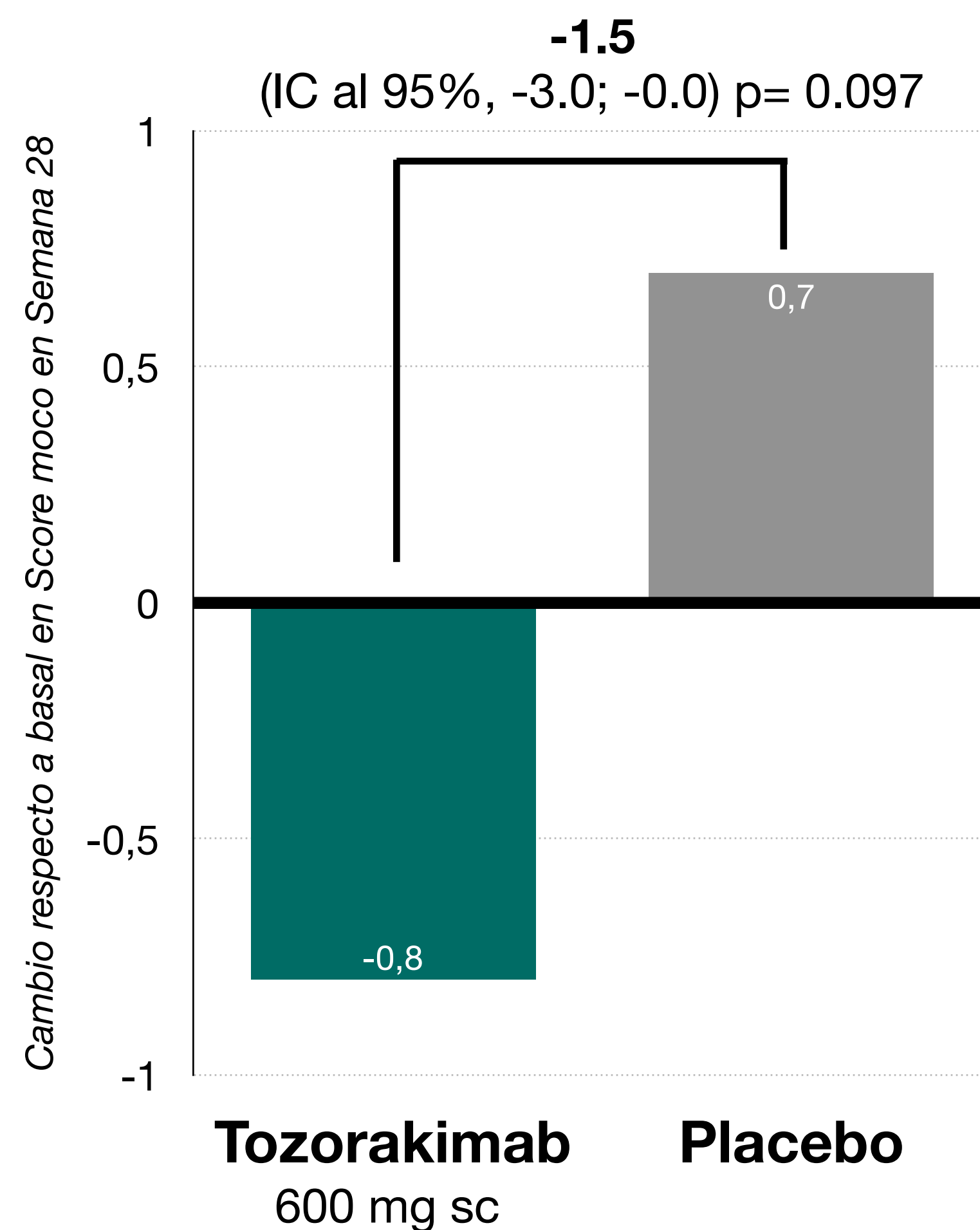
Difference *versus* placebo in the time-to-first **COPDCompEx** event

ITT population ↓ HR: 0.79
80% CI 0.57–1.11;
p=0.186

Patients with at least two prior exacerbations ↓ HR: 0.61
80% CI 0.37–1.00;
p=0.098

- The primary end-point was not met in the ITT population
- Tozorakimab showed promising efficacy signals in patients with a high risk of exacerbation

Tozorakimab (anti- IL33)



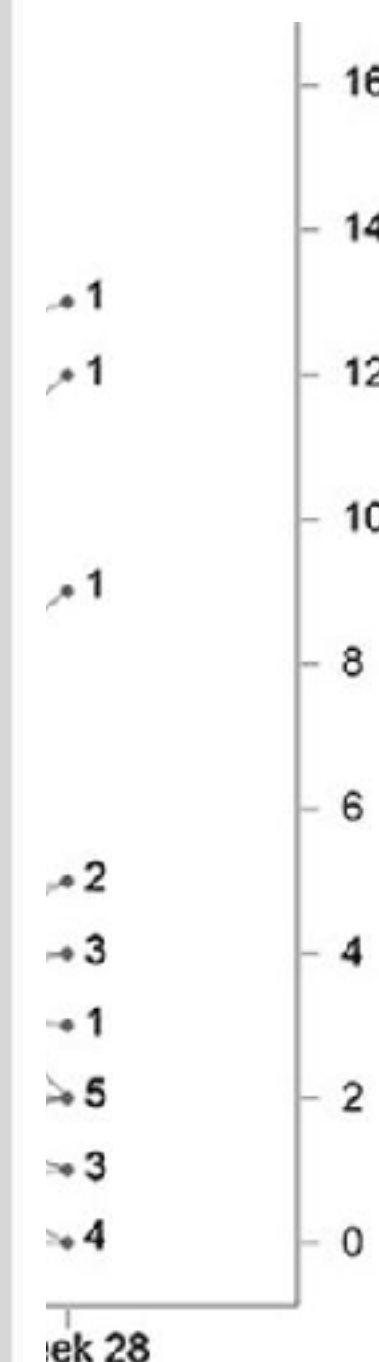
The change in mucus plugging in RB3 airway in a patient treated with tozorakimab
Example CT images



Baseline



Week 28





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Tozorakimab (anti- IL33)



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Tozorakimab met primary endpoint in both OBERON and TITANIA Phase III trials in patients with COPD

PUBLISHED
27 March 2026

This announcement contains inside information

First-ever IL-33-targeting biologic to demonstrate statistically significant and highly clinically meaningful reductions in COPD exacerbations in two replicate Phase III clinical trials

Positive high-level results from the Phase III OBERON and TITANIA trials in patients with chronic obstructive pulmonary disease

<https://www.astrazeneca.com/media-centre/press-releases/2026/tozorakimab-met-primary-endpoint-in-oberon-titania-phase-iii-trials-in-patients-with-copd.html>

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Conclusiones

- Los tapones mucosos parecen un objetivo prometedor como marcador surrogado de eventos futuros en EPOC y por tanto susceptibles de ser modificados farmacológicamente.
- Hasta la fecha, los fármacos inhalados y orales empleados comúnmente en la EPOC sólo han demostrado algunos beneficios sobre el moco en modelos animales o cultivos celulares humanos.
- Dupilumab ha demostrado en pacientes con asma bronquial reducción de tapones de moco que se traduce en beneficios clínicos. Si esto es aplicable en EPOC lo verá el estudio AEOLUS.
- Tozorakimab ha demostrado reducciones en la puntuación de tapones mucosos en pacientes con EPOC en el estudio FRONTIER. A la espera de resultados de OBERON/ TITANIA parece un fármaco a considerar en este aspecto



*Precision medicine begins with
seeing the right enemy*

